

Assignment:- 6A

4-Bit Asynchronous UP Counter using 74LS76

An up counter is a type of digital counter that increments its count by one for each clock pulse it receives. It counts in an ascending sequence, starting from 0 and progressing towards its maximum possible value. For example, a 3-bit up counter would count from 000 (0) to 111 (7) and then reset to 000 to start again.

A 4-Bit Asynchronous UP Counter is a digital circuit that counts upwards from 0000 (0 in decimal) to 1111 (15 in decimal). It is built using four flip-flops (typically T-type or JK-type in toggle mode).

- Asynchronous: The clock input is applied only to the first flip-flop. The output of this flip-flop acts as the clock for the second, the output of the second acts as the clock for the third, and so on. This "ripple" clocking is what defines it as asynchronous.
- 4-Bit: It uses four flip-flops to represent four bits of data, allowing it to count up to 15.
- UP Counter: The sequence of states increments by one for each clock pulse applied to the first flip-flop.

